Hypothesis-driven product development

Lecture 6, IT Product Development

Innopolis University, 2026

Agenda

- Hypothesis-driven entrepreneurship
- Product metrics
- Product analytics
- Observability

Hypothesis-driven entrepreneurship

Quiz

- Which problem are you solving for yourself by building a product in the ITPD course?
- Which problem do entrepreneurs solve for themselves by building products?
- Why can you afford yourself not to try other product ideas in the ITPD course?

Hypothesis-driven entrepreneurship

The goal is to fail fast and to not waste resources on the wrong direction.

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From Hypothesis-Driven Entrepreneurship:

> “Lean startups” use hypothesis-driven approach to evaluate entreprenerial opportunities.
>
> They state falsifiable hypotheses about their business model and test them using MVPs.
>
> Based on test feedback (quantitative and qualitative data), they decide whether to **keep** the business model, to **pivot** by updating the model, or to **abandon** the model.

Example: hypothesis about the business model

Willingness to pay (viability):

- **Belief**: We believe coaches will pay 500 RUB a month to stop scheduling meetings by hand.
- **Test**: paid online meeting booking service pilot offered to the 20 coaches who finish onboarding.
- **Falsified if**: fewer than 3 accept the pilot, or more than half cancel within four weeks.

Lean startup cycle

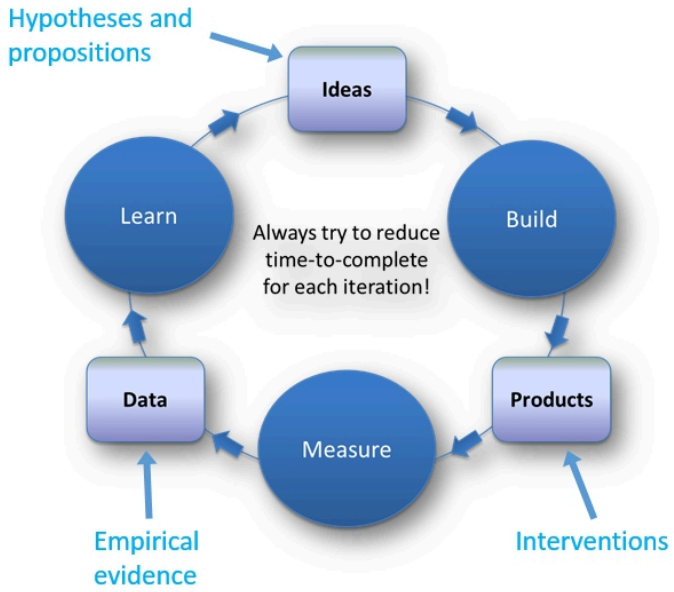


Figure 1: Ries' Lean startup, source

Key takeaways (Hypothesis-driven entrepreneurship)

- Make testable hypotheses so that you can set up experiments.
- Plan experiments and learn from them, don’t just try things.

Product metrics

Quiz

- Which metrics should you care about in your product?

Metric tree

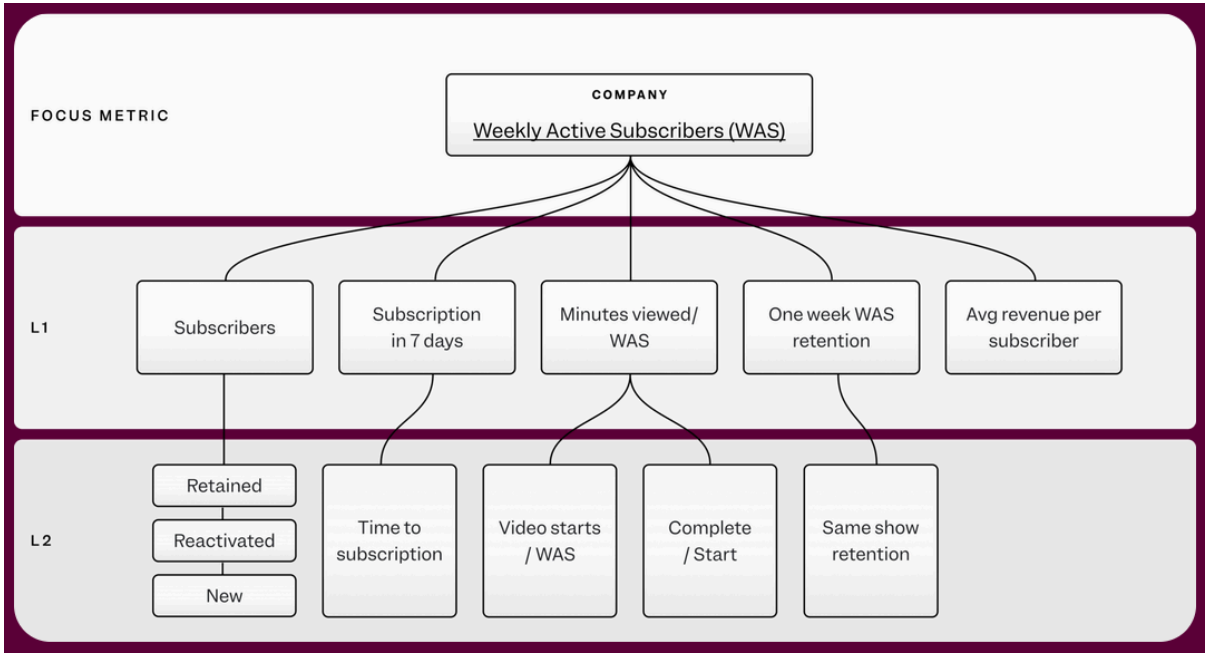
From What is a metric tree:

“Focus Metric” - key actionable metric, a.k.a. “North Star metric”

L1 input metrics feed into the focus metric.

L2 input metrics feed into the L1 metrics.

Metrics can be selected via GQM.



Some types of metrics

From Lean startup:

- **Actionable metrics** - can lean to informed business decisions and subsequent actions.
- **Vanity metrics** - give “the rosiest picture possible” but don’t reflect key business drivers. Opposite of actionable metrics.

From Counter metrics:

- A **counter metric** is what you watch to ensure gains on the north star didn’t come at the expense of customers or the business.

  Example (online meeting booking):

  - North star: completed paid bookings per active coach per week.
  - Counter metric: cancellation and no-show rate within 7 days of booking.

Key takeaways (Product metrics)

- Focus on one key actionable product metric and connect other metrics to it.
- Introduce counter-metrics to avoid gaming the key metric.

Product analytics

Quiz

- How to find out which features are unnecessary in your product?
- How to discover where users behave differently from what you expected?
- How to identify which of your expectations about the user experience are wrong?

Product analytics

From What every product manager needs to know about product analytics:

- Product analytics tracks and analyzes user engagement and behavior (e.g. clicks) to inform product decisions and improvements.
- Data-driven insights help teams understand feature usage, prioritize changes, and pay off “empathy debt.” (show users that “we hear you”)
- Combining analytics with qualitative feedback provides a complete picture of user needs.
- Instrument your product with analytics and use findings to guide feature development and optimization.

Testing a hypothesis about a product change

Possible process for testing hypotheses - from What every product manager needs to know about product analytics:

- Define a clear hypothesis for a product change.
- Build the cheapest possible implementation of this change, loaded with any analytics events you need, that will allow you to test the hypothesis.
- Deploy the change to a subset of customers in an A/B test.
- Wait for results.
- Do a breakdown of the results, with the help of an analyst in the case of more complex changes, and decide whether the change was successful.

Example: hypothesis about a product change

Willingness to book (engagement):

- **Belief**: We believe that making the “Book a meeting” button larger on a coach’s availability page will raise completed bookings by 5% relative, because visitors currently miss the small link among the profile text.
- **Test**: A/B test for two weeks: half the visitors to coach pages see the current link, half see the larger button; compare completed bookings per visitor, with analytics events on page view, slot selected, and booking confirmed.
- **Falsified if**: the 95% confidence interval for the lift lies entirely below +5% relative; if it includes zero, the test is inconclusive and needs more traffic, not a verdict.

Key takeaways (Product analytics)

From What every product manager needs to know about product analytics:

- Think in advance about all events in your system that you need to capture before running experiments.
- Instrument your system to capture as many such events as possible automatically.
- Make dry runs with dummy data to identify gaps.
- Check that your hypothesis is testable given your data and tooling.
- Make sure you’re testing on enough users and for a long enough period of time. You want to make sure your results are statistically significant.
- Be prepared to discard bad ideas quickly.

SLA, SLO, SLI

Quiz

- How do you ensure that your customer doesn’t expect more than what you agreed on?

Why need to document quality requirements?

Problem: mismatch in expectations about desired quality during the project.

Adapted from Non-functional requirements for performance testing:

> When quality requirements are not documented, the system still has characteristics. They're just unknown, untested, and discovered by your customers in production.

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Solution: put quality requirements into a contract.

Adapted from What are SLOs, SLIs, and SLAs?:

- **Service level agreements (SLA)** - legal contracts that explain what is agreed upon and what happens if the system doesn't meet SLOs.
- **Service level objectives (SLO)** - the desired (ranges of) values for specific SLIs that will allow to satisfy SLAs.
- **Service level indicators (SLI)** - key metrics for a characteristic of the system.

Example: SLO, SLI

From What are SLOs, SLIs, and SLAs:

Latency SLOs:

- 95% of web page loads complete within 2 seconds.
- 99% of API requests return within 300 milliseconds.

Latency SLIs:

- Time taken for an API request to return a response.
- Time taken for a web page to load for the end user.

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Note: these SLOs are less precise than quality attribute scenarios because they don’t specify the source of stimulus, stimulus, environment (QAS were explained in Lecture 4).

Key takeaways (SLA, SLO, SLI)

- Formalize agreements about quality (document or even put into a contract)
- Decide in advance what to measure, why, and how

Observability

Quiz

- Which data can help understand why the system malfunctions since the start of this week?

Observability

From Observability pillars:

- **Metrics** - provide quantitative insights into system performance by measuring various parameters (e.g. memory usage, response times, availability). Help **identify that an issue happened** (monitoring).
- **Logs** - provide immutable, exhaustive records of discrete events that occur within a system (e.g. errors). Provide **details to understand the root cause of an issue**.
- **Traces** - capture end-to-end journey of a request through the system, capturing the path and lifespan of each component involved in processing the request. Help **identify sources of issues**.

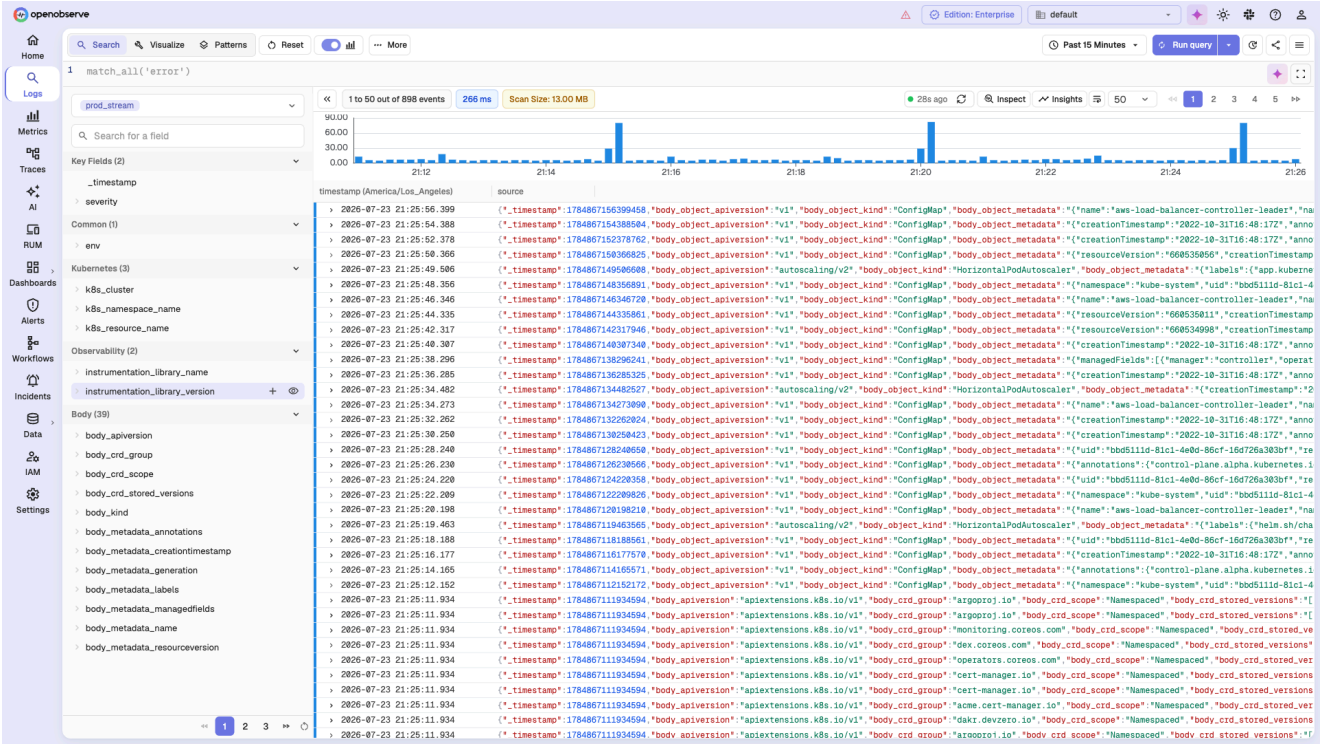
From Instrumentation:

Instrumentation - making code emit signals (metrics, traces, logs).

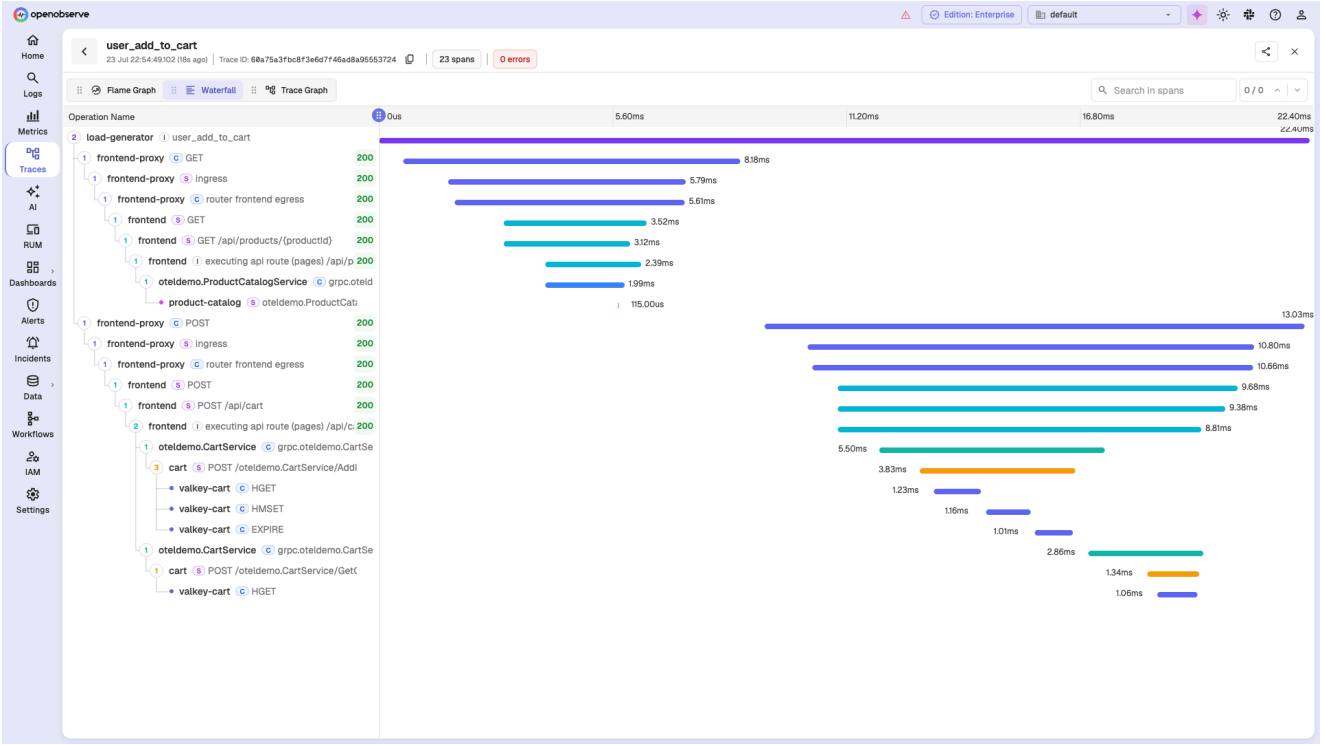
Example: metrics in OpenObserve



Example: logs in OpenObserve



Example: traces in OpenObserve



What does observability enable or improve?

- **Monitoring** - measure SLIs to see how close you are to achieving SLOs and whether SLIs are in a danger zone
- **Alerting** - get notifications when an SLI is in a danger zone
- **Debugging** - observe errors in logs, make hypotheses about their root causes, test them based on collected data
- **Analytics** - capture relevant events
- **SLAs conformance** - demonstrate that relevant SLOs are achieved

Key takeaways (Observability)

- Observability focuses on system health and incident resolution
- Product analytics focuses on product success
- Both observability and analytics require product instrumentation to collect data about events
- Both enable hypothesis testing to resolve issues and make decisions